InCommand Mitiga

Disaster & Incident Command Platform

Capabilities Overview

A single command system for emergency services and disaster management: real-time situational awareness, communications, personnel and resources, field intelligence and ICS documentation, in one place, online or off.

ONE PLATFORM. THE WHOLE INCIDENT.

This overview presents the platform's capabilities for emergency and disaster command.
Screenshots throughout are captured from a live, working prototype.

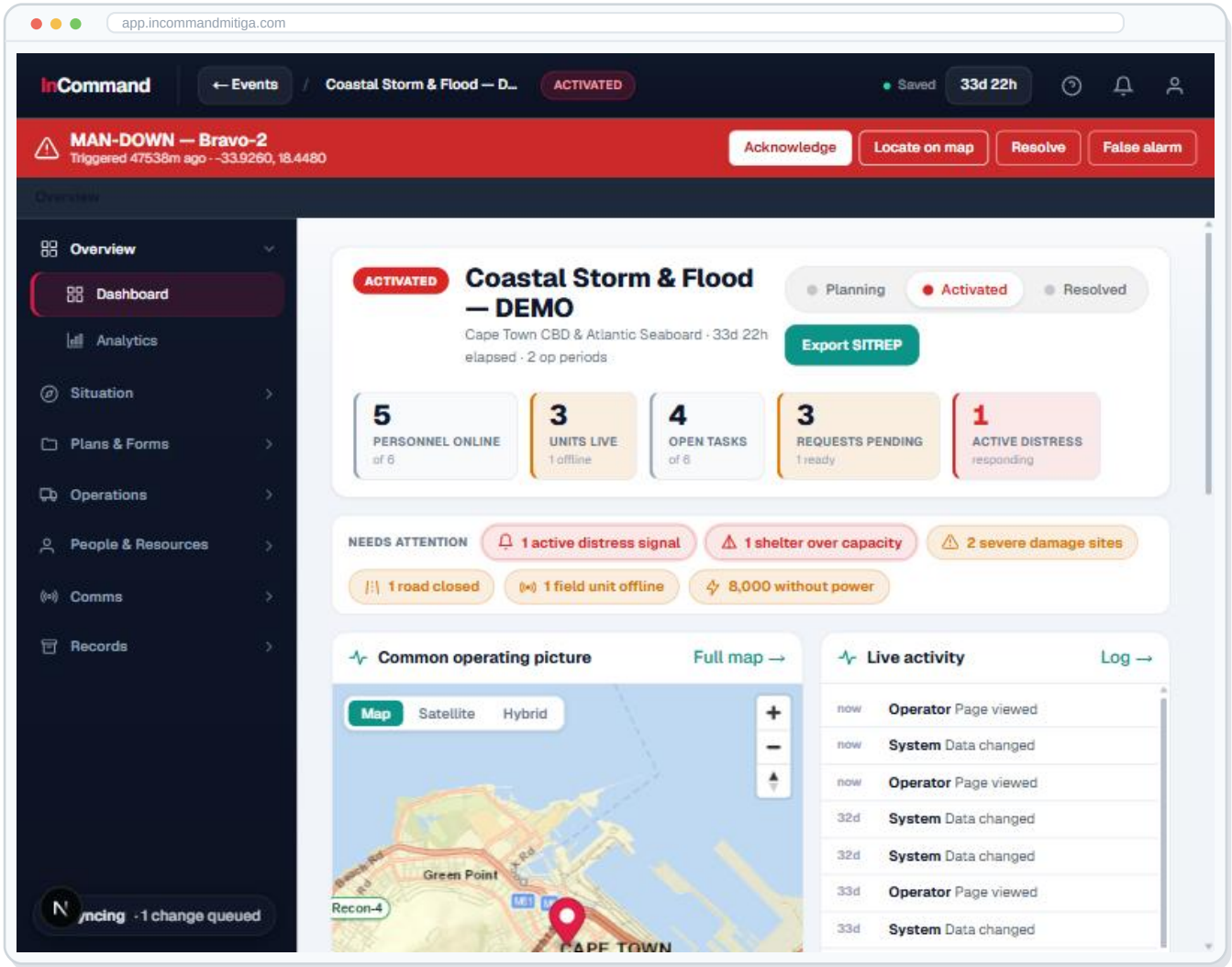
What InCommand Mitiga Is

InCommand Mitiga is a unified command platform for emergency services and disaster management. It gives an incident commander one live picture of everything an incident touches (the situation and map, communications and field units, personnel readiness, resources and equipment, damage and population impact, and the ICS paperwork), instead of a scatter of spreadsheets, radios and whiteboards.

It is multi-tenant (each agency's data is isolated), local-first (it keeps working when the network drops), and real-time (units, video and status update live). It runs in the browser, as a desktop app, and installs on a tablet.

At a glance

- One operating picture: every module feeds a single live command dashboard.
- Any industry, any risk: interoperable across fire, EMS, police, hazmat, hospital and emergency management, single-agency or multi-agency.
- Built around ICS: roles, forms (201/202/205…) and command posture map to how incident command actually works.
- Field-aware: live GPS units, bodycam video, distress/SOS, damage and outage plotting on the map.
- Resilient: offline-tolerant, self-healing live connections, and error capture built in.
- Secure by design: tenant isolation, role-based access, MFA, audit trail.



Live command dashboard: the demo incident “Coastal Storm & Flood”, showing status, needs-attention, the common operating picture and live activity.

Platform Foundation

Cross-cutting capabilities every module is built on:

Architecture

- Multi-tenant SaaS: row-level security isolates each organisation's data at the database.
- Local-first & offline: data lives on the device (IndexedDB) and syncs to the server; the workspace opens and runs even with the backend unreachable.
- Real-time: WebSocket telemetry, paging and signaling with auto-reconnect; live video over WebRTC.
- Cross-device sync: incidents, roster and attendance converge across every device and browser.
- Cross-platform: web app, Electron desktop app, and installable tablet (PWA) from one codebase.

One connected system

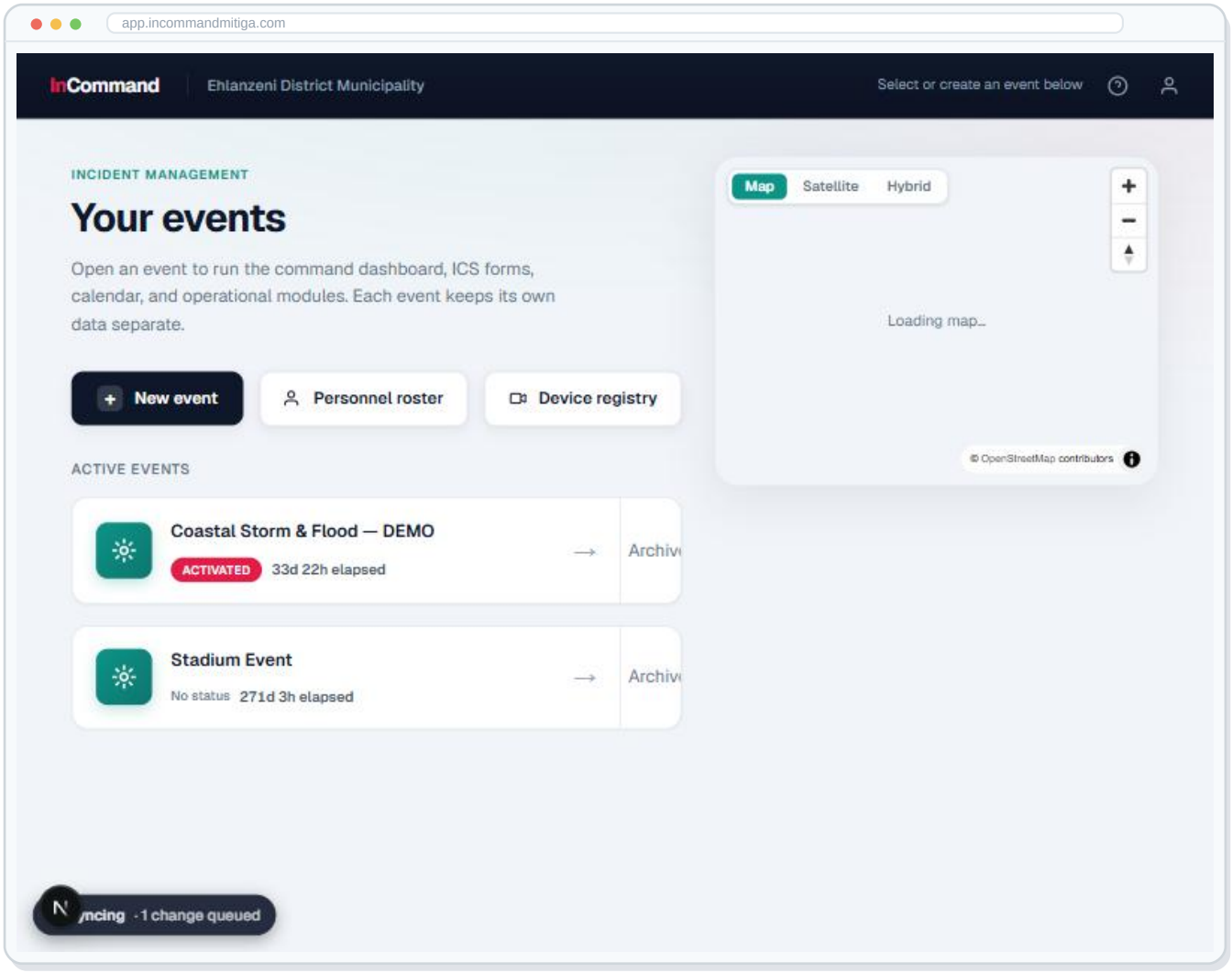
Command dashboards, field bodycams, PTT radios, rugged tablets and vehicle trackers all run on a single platform, so every screen and every device sees the same live picture.



One connected system: command dashboards, field bodycams, PTT radios and rugged field tablets on a single platform.

Multi-incident management

Each incident is fully isolated with its own data, modules, ICS forms and map. Commanders open, run, and archive incidents from a single events hub, branded per organisation.



Events hub: multiple incidents per organisation, each self-contained.

Risk Management & Pre-Planning

Command does not start when the call comes in. InCommand Mitiga's risk tools let agencies map hazards, assess risk and pre-plan the response before an incident ever happens: any industry, any risk, on internationally-accepted ICS foundations.

Capabilities

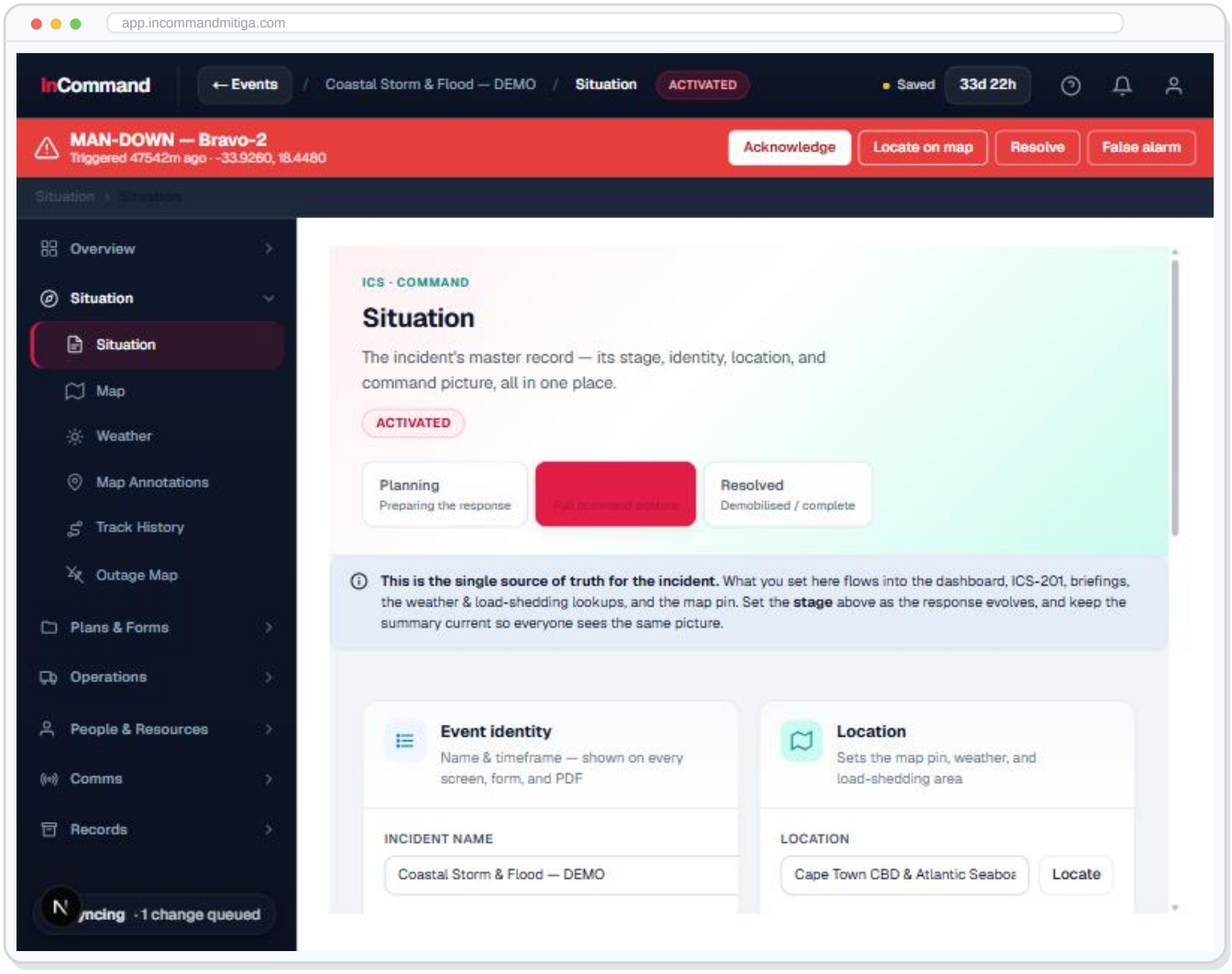
- GIS-based risk mapping: map hazard and risk locations geographically, with data gathered through structured risk-assessment tools.
- Pre-planning: build incident action plans for known hazards and events in advance, ready to activate the moment they occur.
- Mitigation planning: turn assessed risks into mitigation plans, linked to the incidents they inform.
- Templated ICS plans: create single-agency and multi-agency response plans from templated ICS forms.
- Cross-boundary sharing: share risk maps and plans with other institutions and stakeholders on the fly, across organisational boundaries.
- Interoperable across services: ICS-based incident management for fire, EMS, police, hazmat, hospital and emergency management.

Situational Awareness & Command

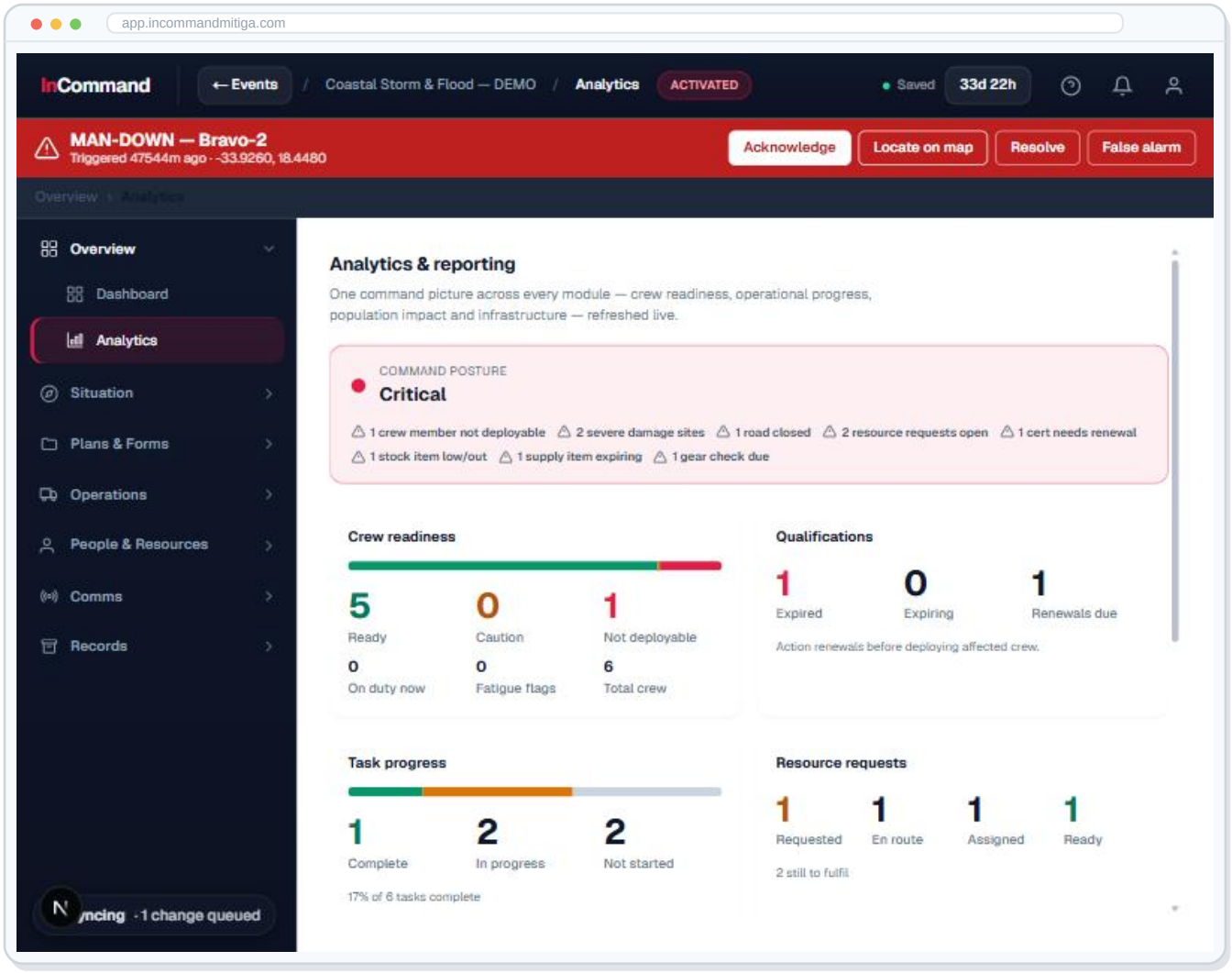
The commander's single source of truth: incident identity, stage, location and the whole command picture in one place, with live analytics on top.

Capabilities

- Situation record: incident name, activation stage (Planning / Activated / Resolved), location and summary that flow into every screen, form and report.
- Command posture analytics: a live, cross-module read of readiness, progress, impact and infrastructure (e.g. “Critical” with the reasons why).
- Weather & environment: meteogram and conditions for the incident location.
- Load-shedding / utility status: regional power status for the affected area (South Africa-aware).



Situation: the incident's master record and stage.



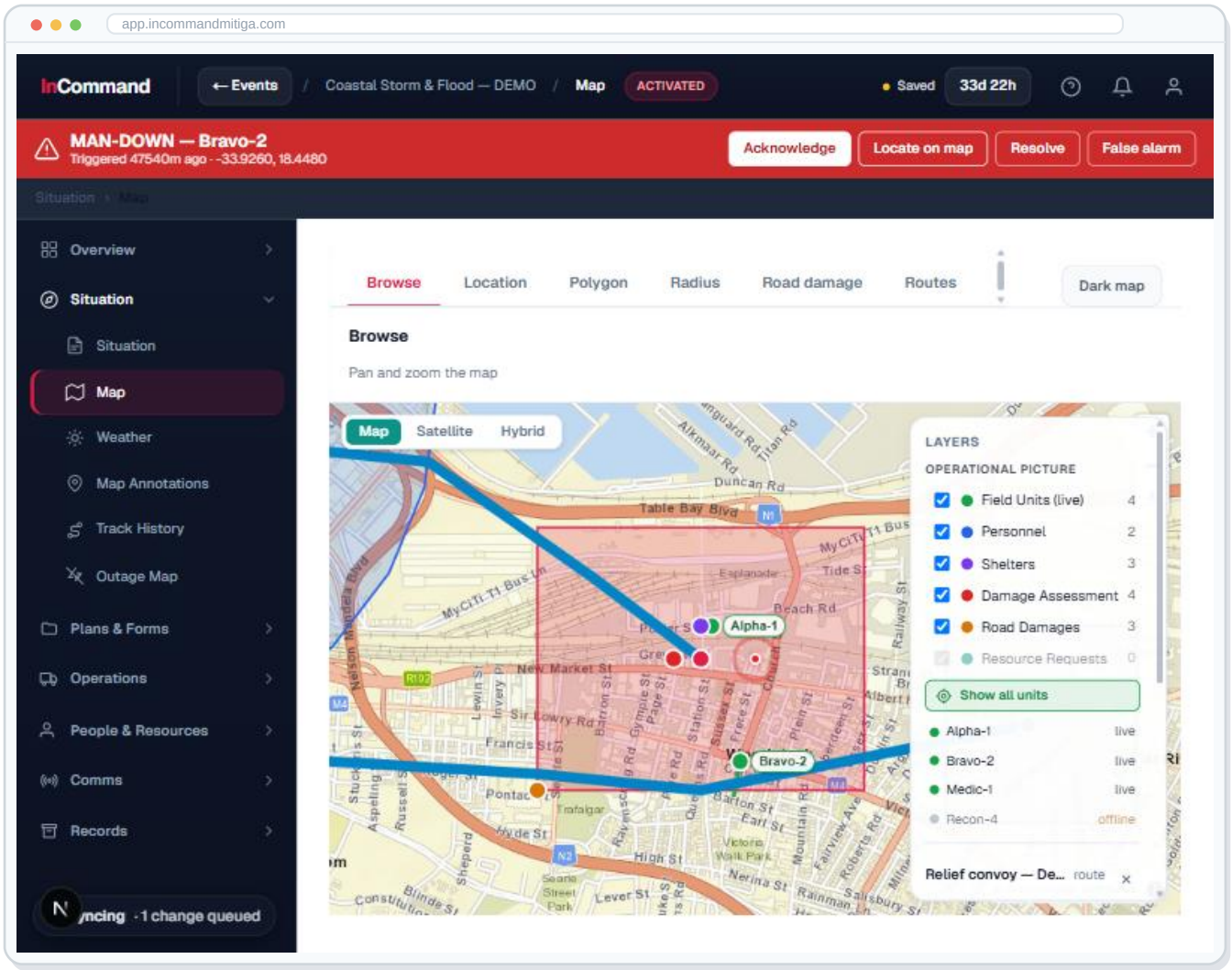
Analytics: command posture, crew readiness, qualifications, task and resource progress, refreshed live.

Common Operating Picture (Mapping)

A live map that fuses everything geographic (units, personnel, damage, hazards, shelters and supply routes) with drawing and routing tools.

Capabilities

- Layered map: street, satellite and hybrid imagery, toggled per operator.
- Live field units: GPS positions, headings, trails and online/offline state on the map.
- Drawing tools: mark points, polygons and radius zones (incident areas, cordons, hazards).
- Damage & hazard plotting: click the map to log damage, road closures and outages with coordinates.
- Supply-route planning: multi-scenario routes (standard, traffic, riot, emergency, diversion) with hazard-avoidance.
- Geocoding & routing: turn an address into coordinates and driving routes (OpenStreetMap base, optional Google enrichment).

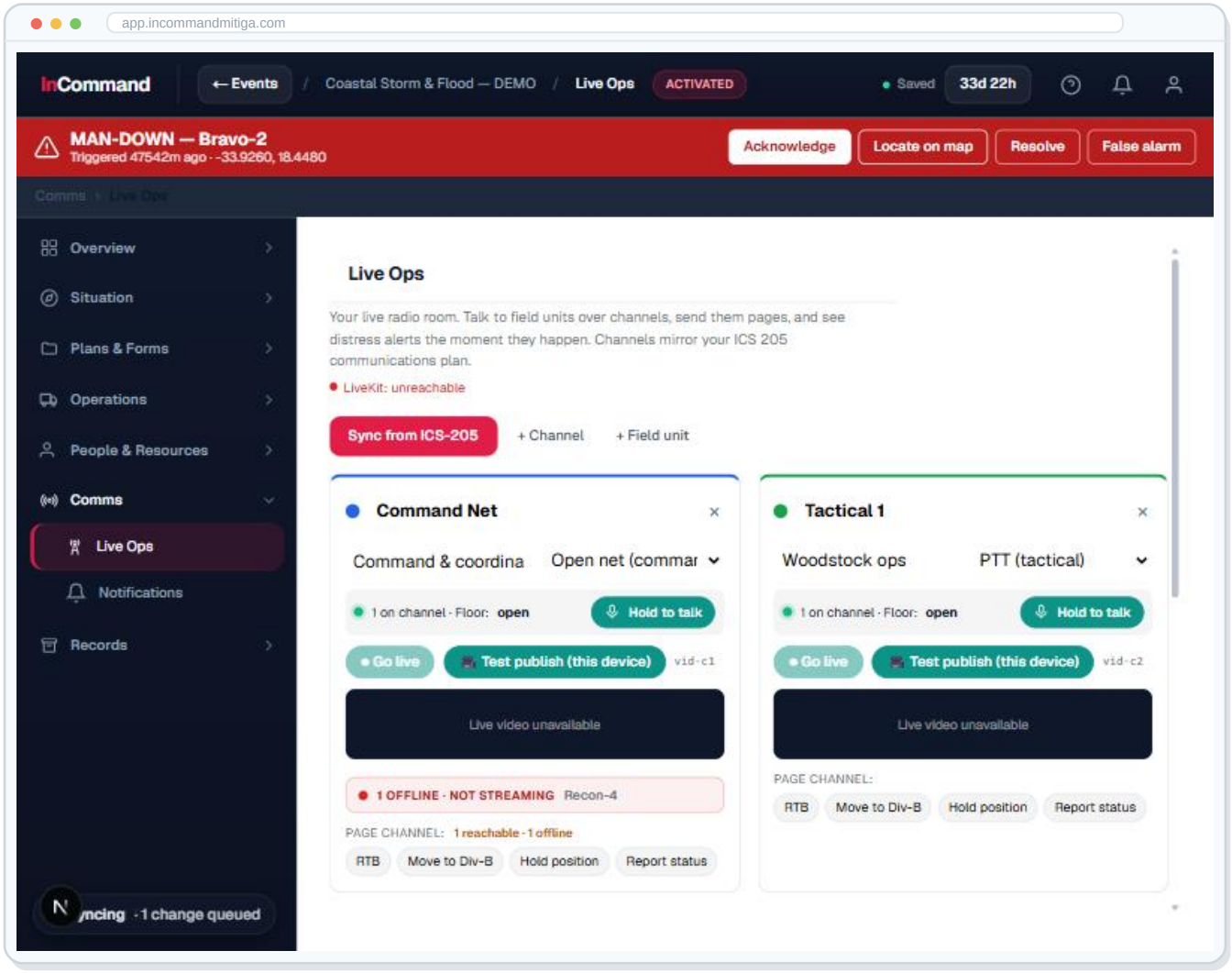

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InCommand
← Events
Coastal Storm & Flood — DEMO
Map
ACTIVATED
Saved
33d 22h
MAN-DOWN — Bravo-2
Triggered 47540m ago · -33.9260, 18.4480
Acknowledge
Locate on map
Resolve
False alarm
Overview
Situation
Situation
Map
Weather
Map Annotations
Track History
Outage Map
Plans & Forms
Operations
People & Resources
Comms
Records
Browse
Location
Polygon
Radius
Road damage
Routes
Dark map
Browse
Pan and zoom the map
Map
Satellite
Hybrid
LAYERS
OPERATIONAL PICTURE
Field Units (live) 4
Personnel 2
Shelters 3
Damage Assessment 4
Road Damages 3
Resource Requests 0
Show all units
Alpha-1 live
Bravo-2 live
Medic-1 live
Recon-4 offline
Relief convoy — De... route
Alpha-1
Bravo-2
Table Bay Blvd
Duncan Rd
Beach Rd
Tide St
Esplanade
New Market St
Sir Lowry Rd
Francis St
Pontac St
Trafalgar
Hyde St
High St
Earl St
Barton St
Nerina St
Lever St
Seans Street Park
Victoria Walk Park
Mountain Rd
Fairview Ave
Roberts Rd
Railway St
Salisbury St
Rainman Ln
Alkmaar Rd
Titan Rd
Nelson Mandela Blvd
MyCiTi T1 Bus Ln
MyCiTi T1 Bus
Chiappini St
Plein St
Sussex St
Station St
Buitenkant St
Bree St
Loop St
Strand St
Albert Rd
Lewin St
Stuart St
Russell St
Aspeling St
Shepherd St
Constitution St
Blinde St
Roeland St
Hanover St
Gympie St
Page St
Barrack St
Queen Victoria St
Ravensberg Rd
Upper Orange St
Upper Mill St
Upper Bree St
Upper Loop St
Upper Buitenkant St
Upper Plein St
Upper Sussex St
Upper Station St
Upper Barrack St
Upper Queen Victoria St
Upper Hanover St
Upper Roeland St
Upper Gympie St
Upper Page St
Upper Ravensberg Rd
Upper Barton St
Upper Earl

Communications & Live Video

A live radio room plus bodycam video: talk to field units, page them, and see distress the moment it happens.

Capabilities

- Channels: command and tactical nets mirrored from the ICS-205 communications plan.
- Push-to-talk: floor control per channel.
- Paging: send pages to units with delivery and acknowledgement receipts.
- Distress / SOS: a takeover banner and map auto-focus the instant a unit raises an alert (e.g. “Man-down”).
- Live bodycam video: units stream to a channel; the operator sees a tile per camera (multi-camera grid).
- Field-ready video: streams over local networks and out to public networks for cellular bodycams in the field.



Live Ops: command & tactical channels, push-to-talk, paging and per-channel video, with a live distress banner.

Call-Taking & Dispatch (Resgrid)

Every incident starts with a call. InCommand Mitiga pairs with Resgrid, an open-source computer-aided dispatch (CAD) platform, so call-taking and unit dispatch flow straight into the command picture.

Capabilities

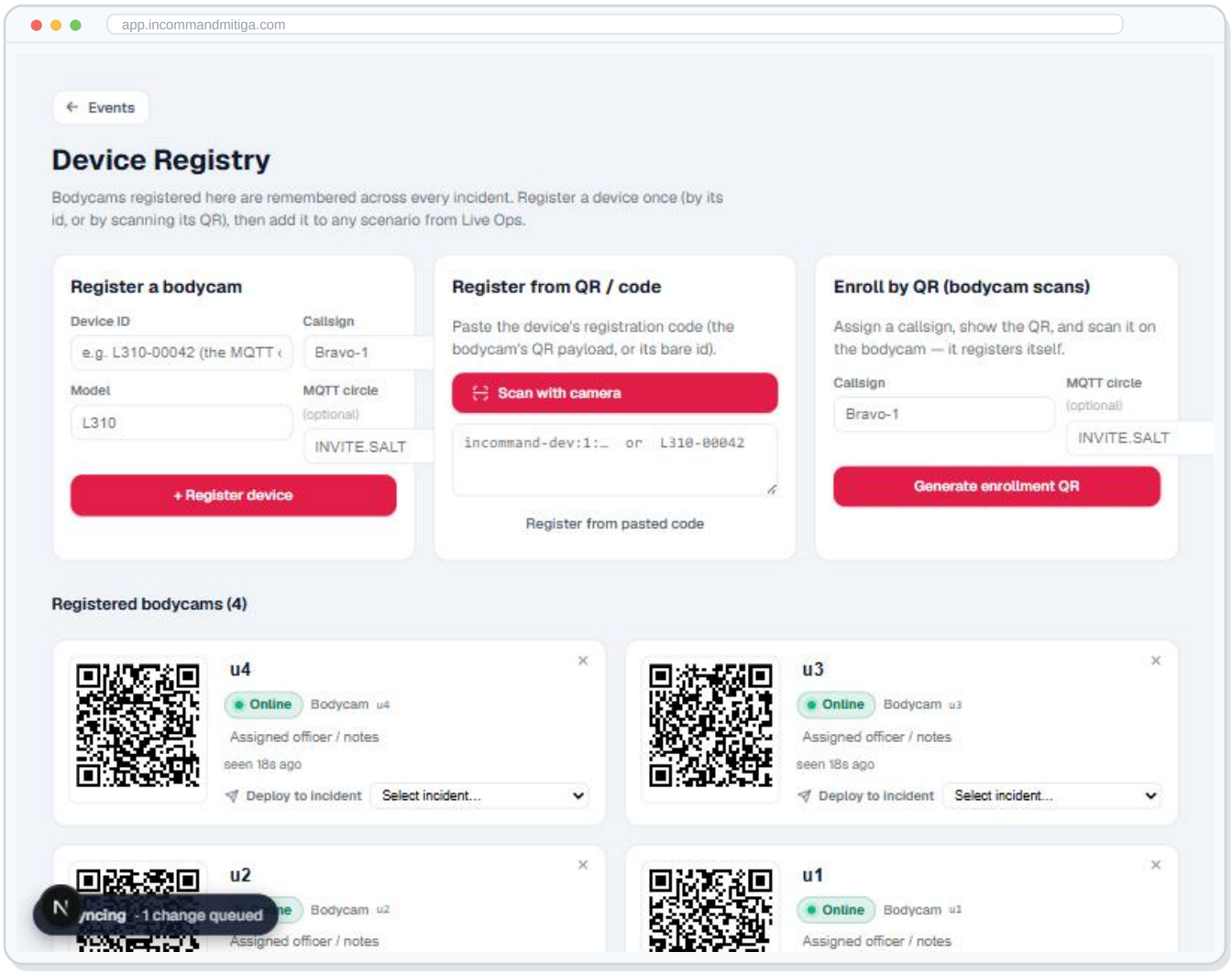
- Call-taking & CAD: create, prioritise, assign and close calls through a full computer-aided dispatch workflow, with multi-channel notifications and analytics.
- Unit dispatching: see unit availability and staffing, dispatch the right resources, and follow the response live.
- AVL & unit control: real-time GPS tracking, 13+ unit statuses, ETA calculation and fleet management for every vehicle and apparatus.
- Staffing & scheduling: shift calendars, self-service signups and shift trades that feed dispatch decisions.
- Multi-channel alerting: notify responders by push, SMS and email, with RSVP tracking.
- Shared operating map: calls, units, personnel and stations on a live map that lines up with the InCommand Mitiga command picture.

Bodycam Fleet & Device Onboarding

An organisation-wide fleet of Mitiga Sentinel FRZ07 bodycams, registered once and reused across every incident (full device specifications included in this overview).

Capabilities

- Device registry: a persistent catalogue of bodycams, separate from any single incident.
- Three-way onboarding: register by ID, paste a registration code, or generate a QR the bodycam scans to enrol itself.
- One-click deploy: drop a registered device into any incident as a live field unit.
- Presence: online/offline status from the device's own check-ins.



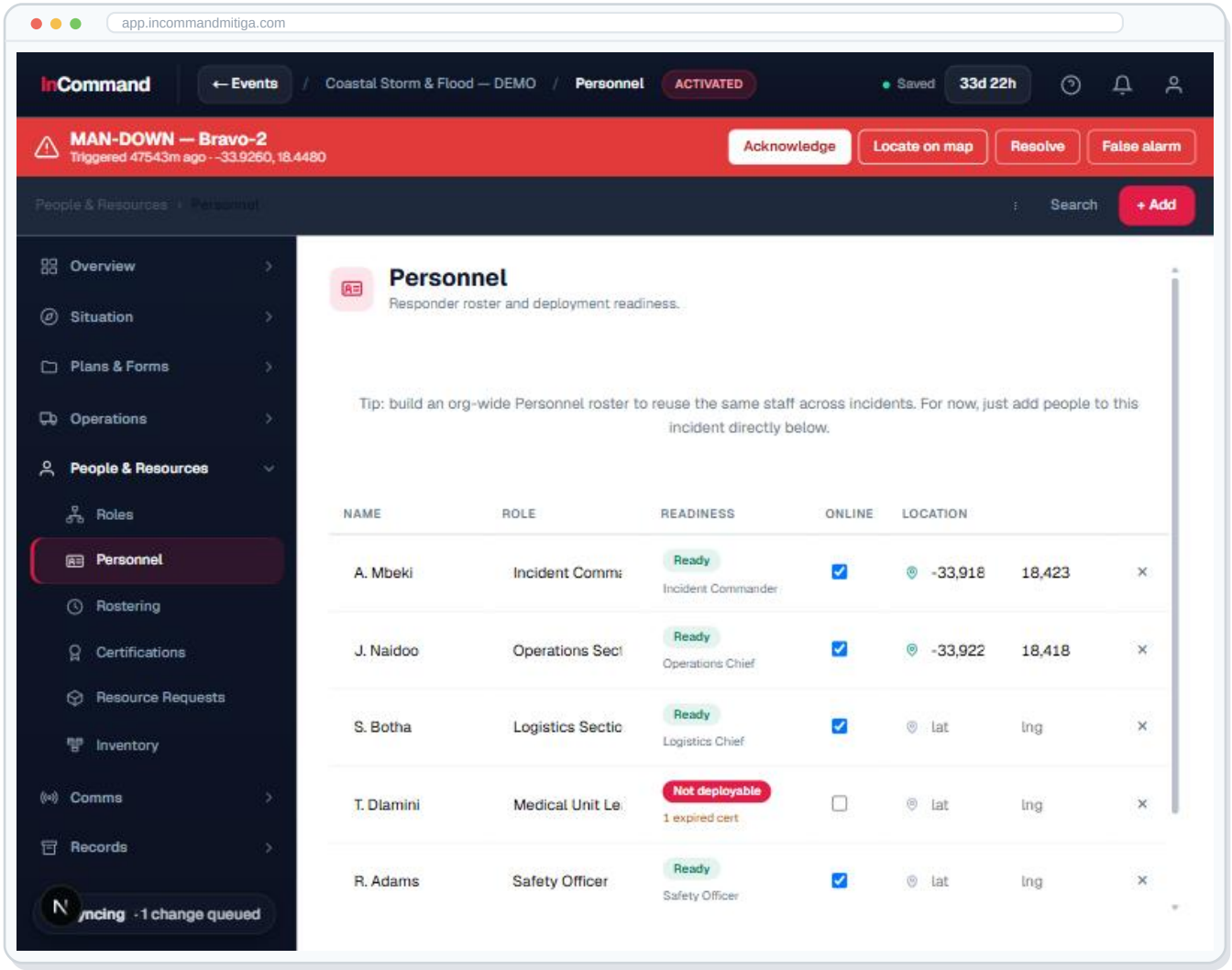
Device registry: QR onboarding and an org-wide bodycam fleet, reusable across incidents.

Personnel, Roles & Readiness

Who's available, who's qualified, who's rested, and whether they can be deployed right now.

Capabilities

- Personnel roster: responders with ICS roles, online status and location.
- Deployment readiness: a live “ready / not deployable” verdict from certifications, fatigue and assignment.
- Roles & ICS structure: role groups and assignments (org chart).
- Rostering & fatigue: shift scheduling with on-duty-hours vs rest-limit tracking so nobody is deployed past a safe threshold.
- Certifications: qualification tracking with expiry (expired certs block deployment).
- Org-wide roster & attendance: a master staff list with experience history and daily attendance, synced across devices.



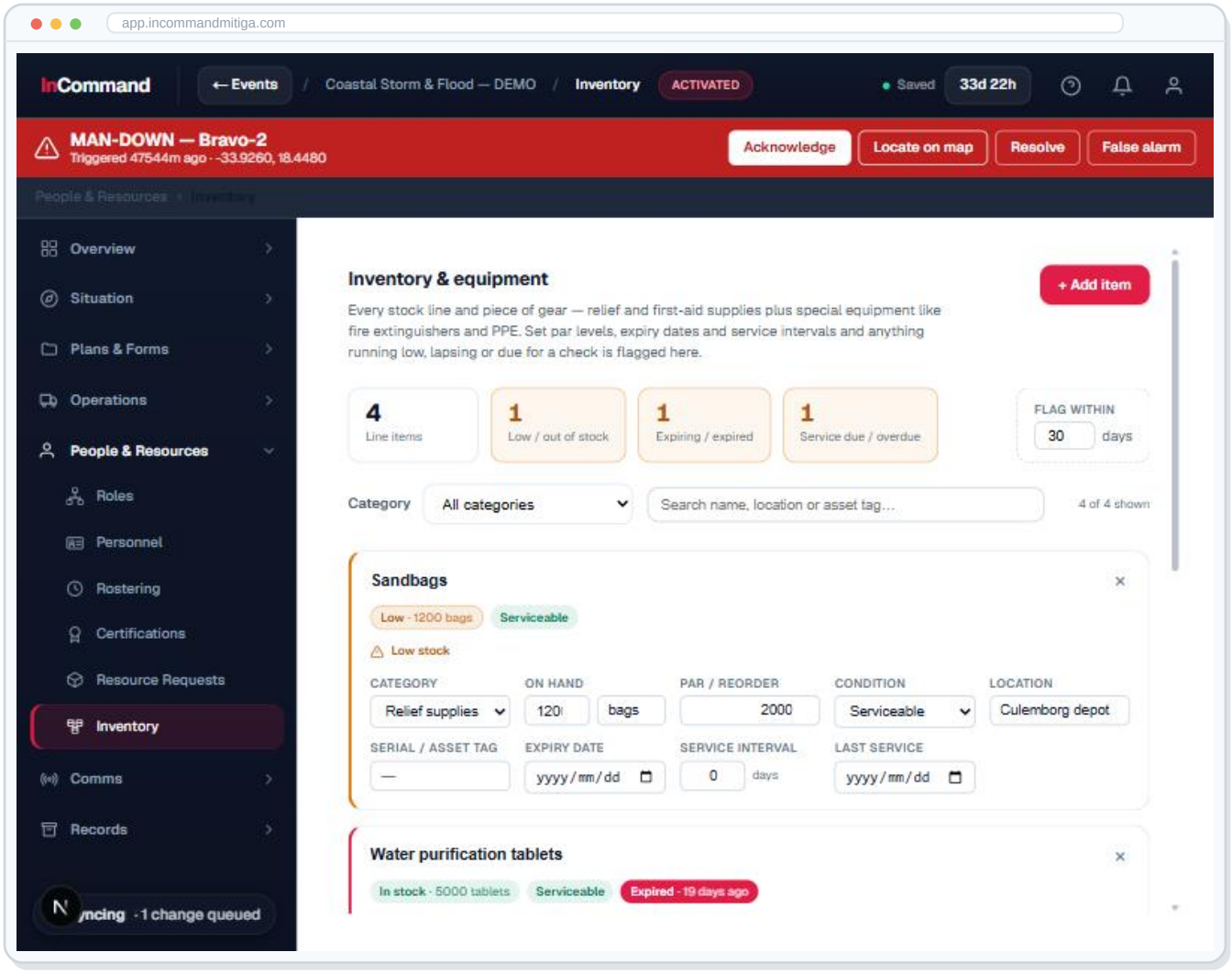
Personnel: responder roster with live deployment-readiness (cert-aware).

Resources & Equipment

Every stock line and piece of gear: what you have, what's running low, and what's due for a check.

Capabilities

- Inventory & equipment: relief/first-aid stock plus special gear (PPE, extinguishers) with par levels, expiry and service intervals.
- Automatic flags: low/out-of-stock, expiring/expired and service-due items surfaced on sight.
- Resource requests: track each request through its lifecycle, from requested to en route, assigned and ready.



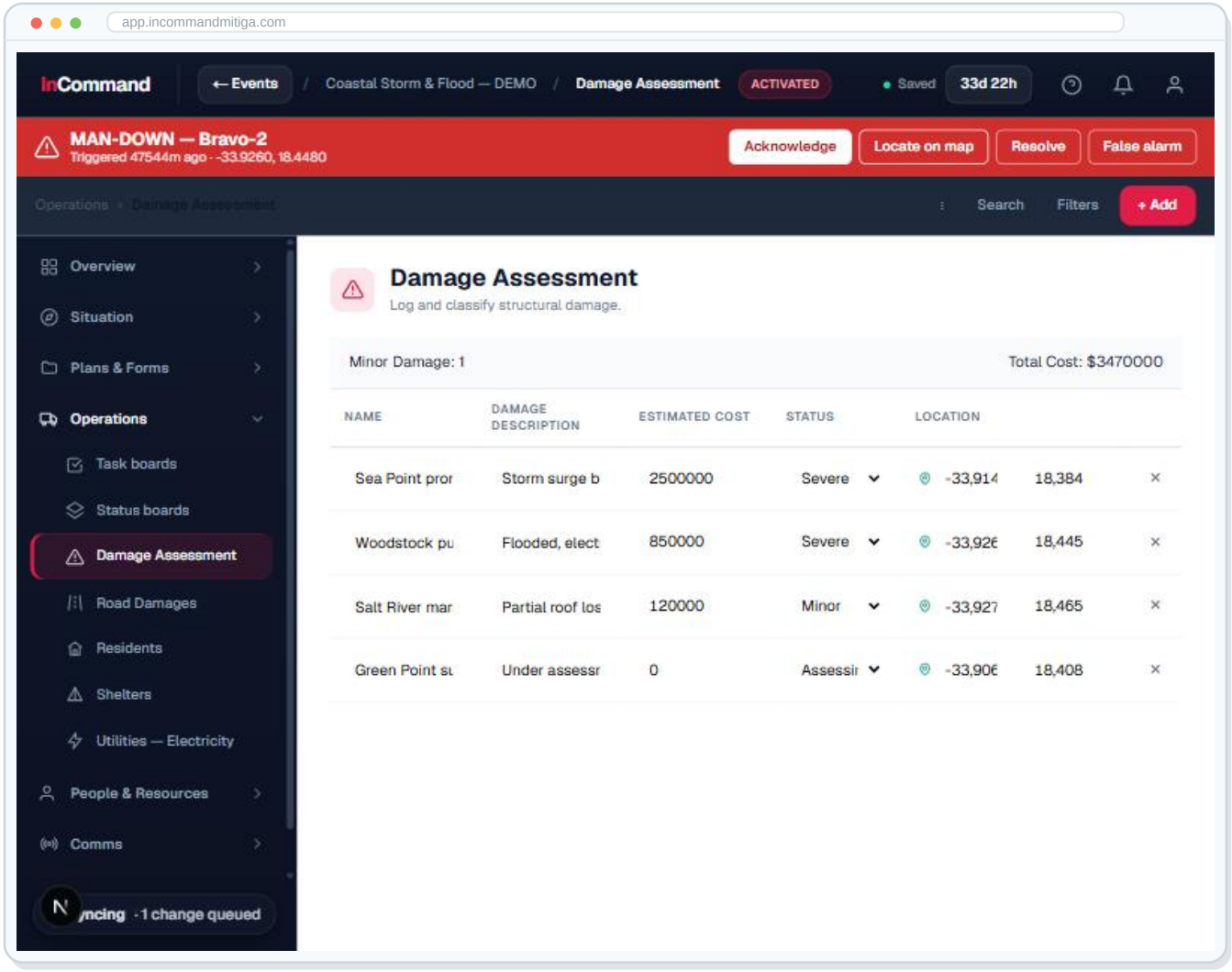
Inventory: stock, expiry and maintenance tracking with automatic low/expired/service-due flags.

Field Intelligence & Population Impact

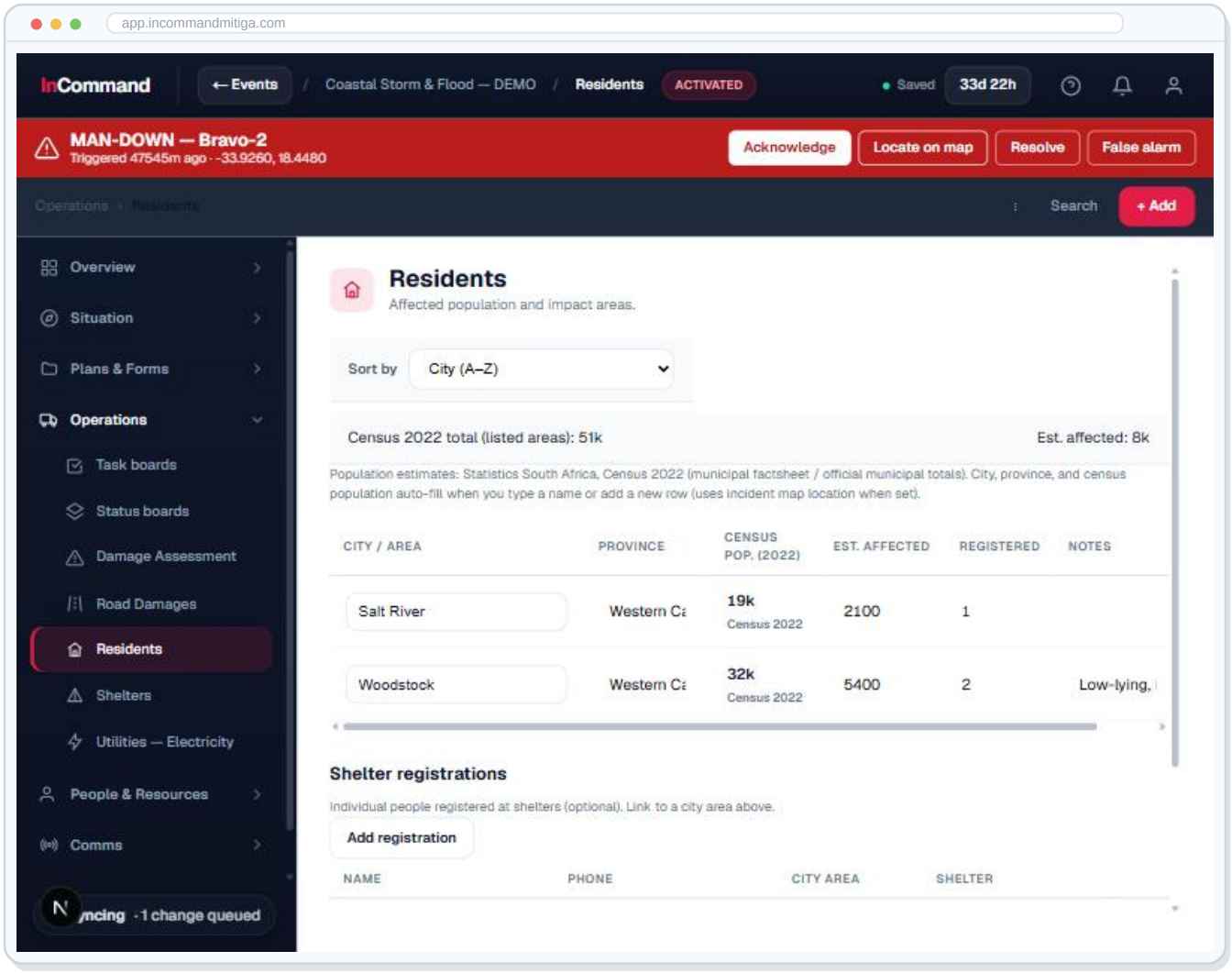
The ground truth: structural damage, infrastructure outages, and the people affected.

Capabilities

- Damage assessment: log and classify structural damage with cost, status and map location.
- Road damages & closures: plot blocked or damaged routes.
- Utilities & outages: power outages and load-shedding status for the affected area.
- Residents & impact areas: affected population by area, auto-filled from census data (Statistics South Africa), with estimated-affected counts.
- Shelters: shelter management and registrations.



Damage assessment: classified structural damage with cost, status and coordinates.



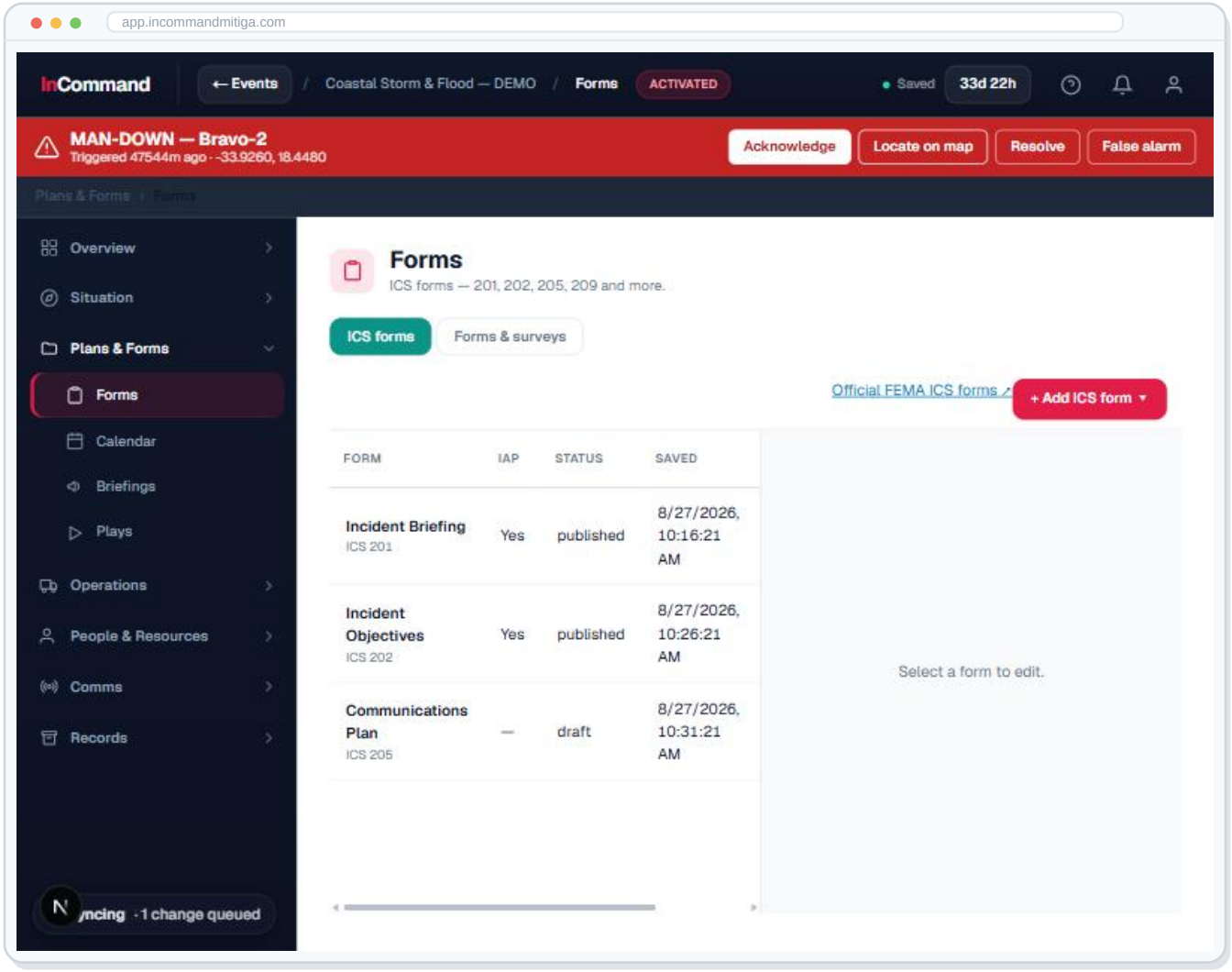
Residents: census-backed affected-population tracking per impact area.

ICS Documentation & Reporting

The paperwork incident command runs on: generated from the live incident, not retyped.

Capabilities

- ICS forms: 201 (Briefing), 202 (Objectives), 205 (Comms Plan) and more, seeded from the situation and kept current.
- Custom form builder: drag-and-drop surveys/checklists with ten field types, filled and collected in-app.
- Document library: live PDF / text exports of forms and records.
- One-click SITREP: a full situation report PDF (map snapshot, weather, status, personnel, forms and tables).
- Logs & audit: activity logging and an audit trail across the incident.



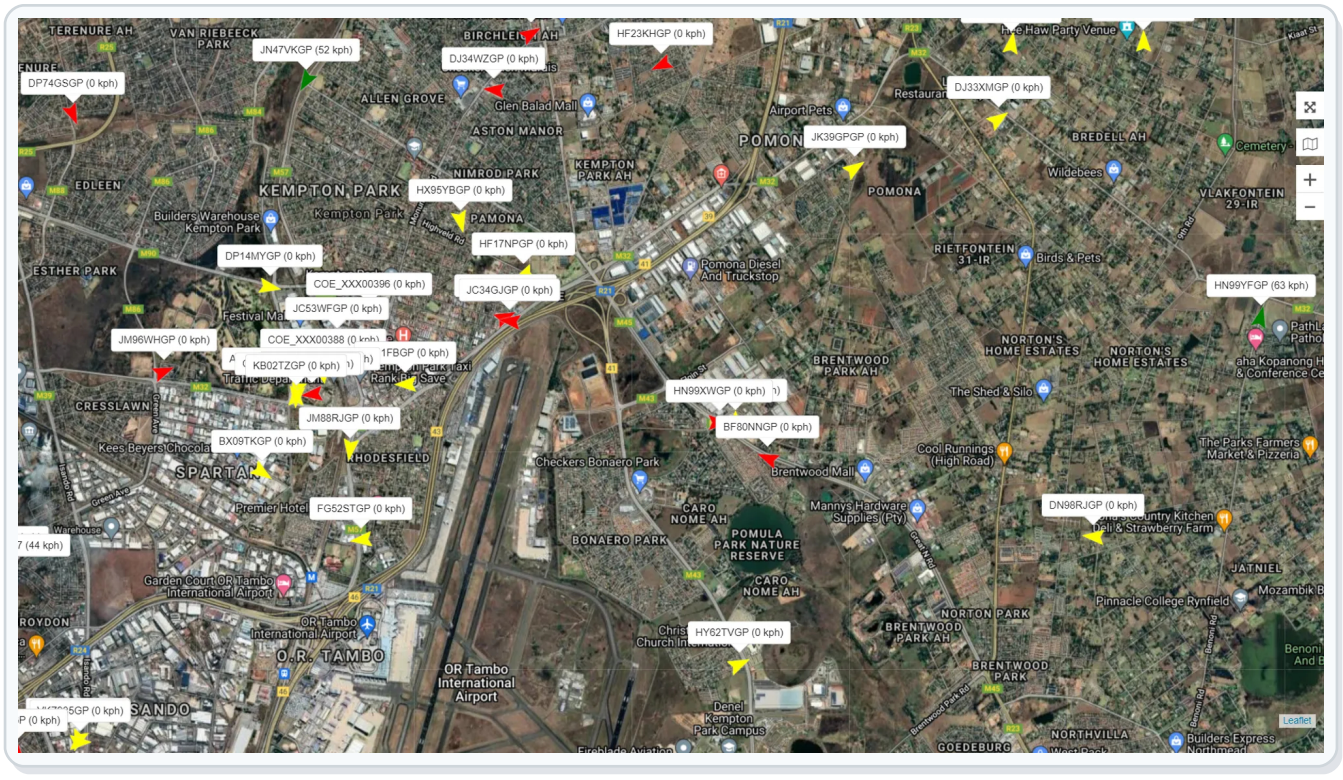
ICS forms: 201 / 202 / 205 generated from the live incident, plus a custom form & survey builder.

Vehicle & Asset Tracking

Command needs to know where its vehicles are and whether they are carrying the right gear. InCommand Mitiga integrates professional Teltonika telematics and GPS Worx fleet tracking to put emergency vehicles and their critical equipment on the same live picture.

Emergency Vehicle Tracking

- Teltonika FMC150 GNSS trackers fitted to fire engines, ambulances and response vehicles for professional-grade, always-on location.
- Live location, speed and status on the GPS Worx fleet platform, feeding straight into the InCommand Mitiga operating picture.
- 4G LTE (with 2G fallback) and multi-constellation GNSS (GPS / GLONASS / Galileo / BeiDou) for reliable positioning in the field.
- CAN-bus and OBDII vehicle data read straight from the vehicle: engine hours, RPM, fuel level, mileage, temperatures and fault codes.
- Driver identification via iButton / RFID, so command knows exactly who is operating each vehicle.
- Accident and crash detection, eco-driving and over-speed scoring, plus ignition, towing and GPS-jamming detection.
- Geofencing, trip history, digital inputs/outputs and over-the-air firmware updates for full fleet oversight.



GPS Worx: live emergency-vehicle tracking across the fleet, feeding the InCommand Mitiga command picture.

Critical Equipment Tracking

- Teltonika Bluetooth tags attached to high-value, life-saving equipment.
- The vehicle's onboard tracker detects the tags aboard, so command knows a Jaws of Life is on the fire engine or a defibrillator is on the ambulance before it rolls.
- Instant flag when critical gear is missing from its assigned vehicle, or left behind on scene.
- One live inventory of the equipment that matters, tied to the vehicle that carries it.

Bodycam Unit: Mitiga Sentinel FRZ07

The InCommand Mitiga field unit is the Sentinel FRZ07: a rugged, two-way-radio-styled Android handset purpose-built for field security, patrol and response, combining a body-worn camera, a duress radio and a connected smartphone in a single ruggedised unit.

Platform & Performance

Operating System	Android (AOSP-based), rugged OEM build
Processor	Multi-core mobile SoC (quad / octa-core)
Memory (RAM)	4 GB
Onboard Storage	32 GB, expandable via microSD
Connectivity	4G / 5G LTE cellular data, Wi-Fi, Bluetooth, GPS / GLONASS

Camera & Imaging

Bodycam Camera	Rear-facing HD video and photo capture
Night Vision	Infrared illuminator with automatic day / night switching
Recording Modes	Continuous, segmented, and hidden / covert capture
Onboard Indicator	Dedicated hardware LED recording indicator

Build & Field Readiness

Form Factor	Two-way-radio-style handheld, chest / lapel mountable
Durability	Rugged sealed housing, drop and dust / water resistant
Controls	Dedicated physical side key for one-touch record / PTT
Battery	High-capacity, all-day field operation
Display	Compact daylight-readable touchscreen

Specifications reflect the Sentinel FRZ07 platform as configured for the InCommand Mitiga deployment; exact battery capacity, imaging resolution and ingress-protection rating are confirmed per production configuration.

Field Unit: Onboard Software

Every Sentinel FRZ07 becomes a connected field unit: live location on a shared map, secure two-way voice and radio-style calling, on-demand live video to command, and automatic evidence capture, all coordinated through the InCommand Mitiga command dashboard.

Live Location & Circle Safety

- Always-on, high-accuracy GPS tracking shared in real time with the unit's assigned circle.
- One-touch SOS, man-down and duress alerts broadcast instantly to command and nearby units.
- Automatic offline / online presence detection, so command always knows which units are reachable.
- Private, invite-key circles keep location and comms scoped to the right group only.

Recording & Evidence Capture

- One-touch video and photo capture from the touchscreen or the dedicated hardware side key.
- Automatic night vision with seamless day / night camera switching.
- Long recordings auto-segment into clean, independently playable clips: no single unbounded file, no lost footage from one corruption.
- Every clip and photo is auto-tagged with timestamp, officer name and GPS location.
- Covert / hidden recording mode for discreet evidence capture.

Live Video Streaming to Command

- Command can remotely trigger any online bodycam to go live and stream its feed in real time.
- Low-bandwidth, publish-on-demand streaming: video only flows when needed, preserving cellular data.
- Multiple bodycams stream simultaneously into a single multi-camera live grid on the dashboard.
- Secure, authenticated publishing: only verified, enrolled devices can stream into a command channel.

Voice & Radio-Style Calling

- Secure 1:1 calling between circle members over encrypted SIP or WebRTC.
- Group and talkgroup push-to-talk, radio-net style, for whole-team coordination.
- Automatic fallback between calling paths so voice stays available as network conditions change.
- Radio-over-IP (RoIP) ready: bridges cleanly into existing two-way radio fleets.

Enrolment & Security

- Scan-to-register: point the bodycam at a QR code on the command dashboard and it self-enrols in seconds.
- Cryptographically signed device authentication (HMAC) for every call, stream and enrolment action.
- End-to-end encrypted media (TLS / SRTP) across voice, video and command-channel traffic.
- Self-hostable backend: run the entire command and calling infrastructure on your own private infrastructure.
- Kiosk-mode lockdown keeps the device fixed to the bodycam application in the field.

Security & Compliance

Built for sensitive incident and personnel data across multiple agencies.

- Tenant isolation: row-level security enforces per-organisation data separation at the database.
- Authentication: organisation (district) login with password, plus MFA / one-time codes.
- Role-based access control: per-module read/write permissions by ICS role.
- Account safety: admin invites, password resets, lockout and rate-limiting, session/JWT revocation, breached-password checks.
- Audit trail: a recorded history of actions across the incident.
- Hardened transport: strict content-security policy and security headers.

Reliability & Resilience

- Offline-tolerant workspace: opens and runs with the backend unreachable; changes queue and sync when it returns.
- Self-healing live connections: telemetry, comms and video reconnect automatically after drops or sleep.
- Error capture: client-side error boundaries isolate a failing panel (the rest of the console keeps working) and report issues so failures are seen, not silent.
- Installable & always-current: PWA install with automatic update on launch.

Platforms & Deployment

Runs everywhere

- Web: any modern browser.
- Desktop: an Electron app (Windows/macOS/Linux), either thin-client to the live server or a self-contained build.
- Tablet & mobile: installs from the browser as a full-screen app (PWA), online or offline.

Infrastructure

- Container stack: Postgres, object storage (S3/MinIO), Redis, live-video server and mail, via Docker.
- Scales out: multi-replica ready, managed-Postgres and Kubernetes-oriented.

Field Deployment & Connectivity

Built to run where incidents happen: on the network you have, on the devices your teams carry, alongside the systems you already run.

- Live field video: bodycam and device video streamed into the command picture, over local networks or cellular.
- Access anywhere: hosted securely for your organisation so commanders, tablets and remote responders connect from anywhere.
- Runs on your terms: on your own infrastructure or in the cloud, with no per-device rebuild.
- Fits your ecosystem: integrates with Resgrid for call-taking and dispatch, and connects with the other systems agencies already run (radios, fleet).
- Always current: the whole command team runs the latest version automatically, no manual updates.

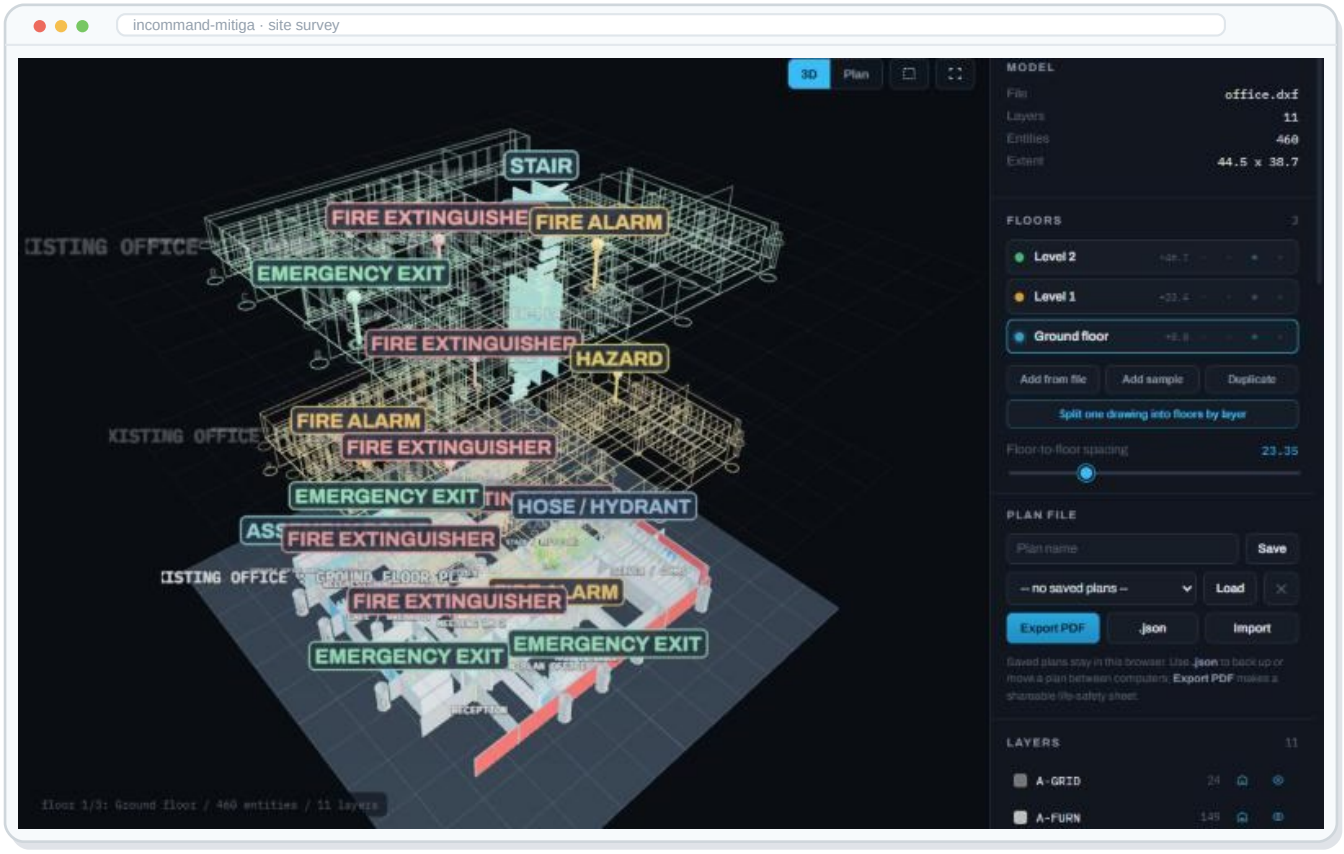
One platform, the whole incident.

From the first alert to the after-action report, InCommand Mitiga brings situation, map, comms, people, resources, impact and paperwork into one live, resilient command picture, replacing the scatter of tools an incident commander juggles.

ON THE ROADMAP · IN DEVELOPMENT

The Future of Fire and Risk Surveys

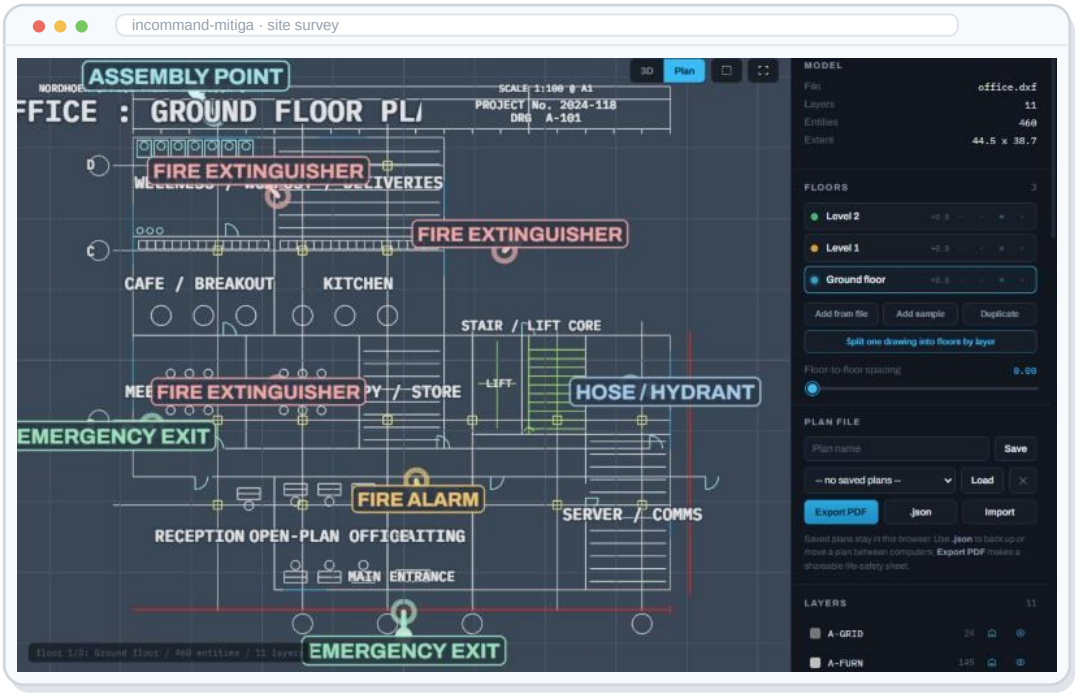
A dedicated site-survey engine, now in development, that plugs straight into InCommand Mitiga. It reads floor and site plans directly from AutoCAD (DXF) files and turns them into live fire and life-safety maps, so the survey done before an incident becomes the plan command uses during it.



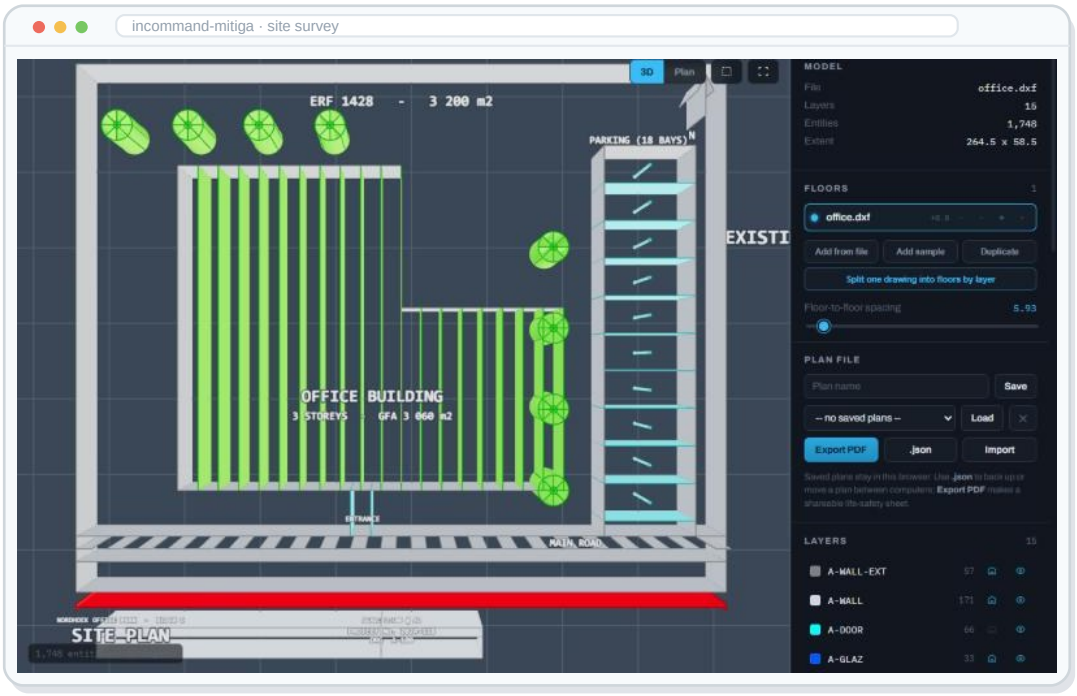
Exploded 3D view of a multi-storey building: every floor and every life-safety marker in one model, read straight from the CAD drawing.

What it does

- Reads AutoCAD directly: opens DXF floor and site plans and pulls them apart layer by layer, floor by floor, with no redrawing.
- Maps life-safety assets: places markers for emergency exits, stairs, assembly points, fire alarms, smoke detectors, extinguishers, hoses and hydrants.
- Flags risks and hazards: pins hazards and risk points to the exact spot on the plan.
- Handles multi-storey buildings: splits one drawing into floors by layer and plans each level on its own (ground, level 1, level 2 and up).
- Plan and 3D views: switches between a flat plan and a full 3D model for briefings, walkthroughs and pre-incident familiarisation.
- Shareable life-safety sheets: exports a marked-up plan as a fire and life-safety PDF for crews and building owners.



Ground-floor plan with life-safety markers on the CAD drawing: assembly point, extinguishers, hydrant, alarms and marked emergency exits.



Site plan in 3D: building footprint, roof, parking and access, for site-wide pre-planning and approach.

Preview of what's coming

This engine is in active development and is not yet part of the live platform. It is shown here as a preview of where InCommand Mitiga is heading: pre-incident risk mitigation and prevention, feeding directly into the command picture responders use on the day.

The Complete Suite

One platform, from risk to recovery. Every capability, connected.

Risk & Pre-Planning

GIS hazard mapping, pre-plans, mitigation

Situation & Analytics

Stage, command posture, weather

Common Operating Picture

Live map: units, zones, routes

Communications & PTT

Channels, paging, distress alerts

Live Bodycam Video

Multi-camera streams to command

Call-Taking & Dispatch

Resgrid CAD integration

Personnel & Readiness

Roster, fatigue, certifications

Resources & Equipment

Inventory, requests, gear

Field Intelligence

Damage, outages, residents

Vehicle & Asset Tracking

Teltonika FMC150 + GPS Worx

Sentinel FRZ07 Devices

Rugged bodycam field units

ICS Docs & Reporting

201 / 202 / 205, SITREP, audit

Plan **Command & Respond** **People & Resources** **Field & Records**

Why InCommand Mitiga

Most agencies run a drawer full of disconnected tools. InCommand Mitiga replaces the scatter with one platform built for the way incidents actually run.

One platform, not ten

Situation, map, comms, video, people, resources, impact and ICS paperwork in a single live picture, not a dozen tabs and radios.

ICS-native

Built around the incident command system your teams already use: forms, roles, status boards and after-action reporting come standard.

Works in the field

Runs on the network you have. Opens offline, queues changes, and self-heals live video, comms and telemetry after drops.

Command to the field edge

Bodycams, vehicles and tagged equipment report onto the same map the commander is already looking at.

Your data, your terms

Tenant-isolated and self-hostable, on your own infrastructure or in the cloud, with a full audit trail.

Ready for what is next

An open, integrating platform (Resgrid, Teltonika, GPS Worx) with a site-survey engine and more on the roadmap.